

SOCIOL 723: Problem Set 1

Assigned: September 1 • *Due:* Monday, September 14, 11:59PM

Your name:

Instructions. Submit a single PDF rendered from a Quarto or RMarkdown source, together with the source file. Show your work: for algebra, the steps; for computation, the code. You may use AI assistance, but you are responsible for understanding and being able to explain every line you submit.

Questions marked ★ are optional and will not be graded. They are there if you want to push on the general versions of the arguments.

The data file is `Data/gss_earnings.rds`, the same extract used in lab.

Least Squares by Hand (20 points)

Consider the bivariate model $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$ estimated by ordinary least squares on a sample of size n .

1. Starting from the objective $S(b_0, b_1) = \sum_i (Y_i - b_0 - b_1 X_i)^2$, derive the two normal equations. State clearly what each one implies about the fitted residuals $\hat{\epsilon}_i$.
2. Show that $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$ and $\hat{\beta}_1 = \sum_i (X_i - \bar{X})(Y_i - \bar{Y}) / \sum_i (X_i - \bar{X})^2$.
3. A colleague fits the model, plots $\hat{\epsilon}_i$ against X_i , observes no correlation, and concludes that the model is correctly specified. In two or three sentences, explain why this reasoning is wrong.
4. Suppose you rescale the regressor to $X_i^* = a + cX_i$ for constants a and $c \neq 0$. Derive what happens to $\hat{\beta}_0$, $\hat{\beta}_1$, the residuals, and R^2 . Use your result to explain what changes and what does not when you center a variable.
5. Show that R^2 from the bivariate regression equals the squared sample correlation between X and Y .

Partialling Out (25 points)

In lecture we showed that the coefficient β_1 in

$$Y_i = \beta_1 D_i + \beta_2' W_i + \epsilon_i$$

can be recovered from a bivariate regression of residualized variables,

$$\tilde{Y}_i = \beta_1 \tilde{D}_i + u_i, \quad \beta_1 = \frac{\text{Cov}(\tilde{Y}_i, \tilde{D}_i)}{\text{Var}(\tilde{D}_i)},$$

where $\tilde{D}_i = D_i - \gamma'_{DW} W_i$ and $\tilde{Y}_i = Y_i - \gamma'_{YW} W_i$ are the residuals from regressing D_i and Y_i on W_i .

1. Show that β_1 can equally be recovered *without* residualizing the outcome:

$$\beta_1 = \frac{\text{Cov}(Y_i, \tilde{D}_i)}{\text{Var}(\tilde{D}_i)}.$$

(Hint: write $Y_i = \tilde{Y}_i + \gamma'_{YW} W_i$ and use the fact that $\tilde{D}_i$ is orthogonal to W_i by construction.)

2. Is the symmetric statement also true? That is, does

$$\beta_1 \stackrel{?}{=} \frac{\text{Cov}(\tilde{Y}_i, D_i)}{\text{Var}(\tilde{D}_i)}?$$

Prove it or give a counterexample.

3. Now consider what happens if you change the **denominator** as well, that is, if you simply regress $\tilde{Y}_i$ on the raw D_i :

$$\frac{\text{Cov}(\tilde{Y}_i, D_i)}{\text{Var}(D_i)}.$$

Show that this does *not* equal β_1 , and show which direction the discrepancy runs. Explain in one sentence what this tells you about which residualization is doing the work.

4. Give an expression for $\text{Var}(\hat{\beta}_1)$ in terms of $\sum_i \tilde{d}_i^2$, where $\tilde{d}_i$ is the residualized regressor. Use it to explain, in plain language, why adding a control that is highly correlated with your treatment inflates the standard error, and why this is *not* a violation of any regression assumption.
5. **Empirically.** Using the GSS extract, verify the Frisch–Waugh–Lovell theorem for the coefficient on `educ` in

```
lm(l_earn ~ educ + exper + I(exper^2) + female + black + wordsum + pareduc,
    data = gss)
```

by residualizing `educ` and `l_earn` on all the other regressors and running the bivariate regression. Report both coefficients and confirm they agree to at least six decimal places. Then produce the added-variable plot and report $sd(\text{educ})$ and $sd(\widetilde{\text{educ}})$.

6. ★ Prove the general matrix statement: with $M_2 = I - X_2(X_2'X_2)^{-1}X_2'$, show that $\hat{\beta}_1 = (X_1'M_2X_1)^{-1}X_1'M_2y$.

Omitted Variable Bias (25 points)

1. Derive the omitted variable bias formula $\tilde{\beta}_1 = \beta_1 + \beta_2\delta_1$ in scalar form, stating clearly where each assumption is used.
2. A researcher regresses log earnings on years of schooling and finds $\hat{\beta}_1 = 0.12$. A referee objects that cognitive ability is omitted. Suppose ability raises earnings and is positively associated with schooling. In which direction is $\hat{\beta}_1$ biased? Would controlling for a *noisy proxy* for ability (such as a 10-word vocabulary test) fully remove the bias? Explain.
3. Now suppose the researcher instead controls for **occupational prestige**, which is plausibly a consequence of schooling. Explain, using the language of good and bad controls, why this is likely to make things worse rather than better. What quantity is the coefficient on schooling now estimating?
4. **Empirically.** Using the GSS extract:
 - a. Estimate the “long” model with `wordsum` and `pareduc` and the “short” model without them.
 - b. Estimate the two auxiliary regressions and verify numerically that $\tilde{\beta}_{\text{educ}} - \beta_{\text{educ}} = \beta_{\text{wordsum}}\delta_{\text{wordsum}} + \beta_{\text{pareduc}}\delta_{\text{pareduc}}$.
 - c. Repeat the exercise adding **prestige** to the long model. Report how the `educ` coefficient moves, and comment on whether you would include it.
5. Suppose two variables are omitted, W_1 and W_2 , with $\beta_{W_1} > 0$, $\delta_{W_1} > 0$, $\beta_{W_2} > 0$, $\delta_{W_2} < 0$. Can you sign the total bias? What would you need to know?

Projection and Residuals (15 points)

Let $\mathbf{P} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ and $\mathbf{M} = \mathbf{I}_n - \mathbf{P}$.

1. Show that $\mathbf{P}$ and $\mathbf{M}$ are symmetric and idempotent, and that $\mathbf{PM} = \mathbf{0}$.
2. Show that $\text{tr}(\mathbf{P}) = k$ and $\text{tr}(\mathbf{M}) = n - k$. Explain in one or two sentences why this is the source of the $n - k$ in $s^2 = \hat{\mathbf{e}}'\hat{\mathbf{e}}/(n - k)$.
3. Using $\hat{\mathbf{e}} = \mathbf{M}\boldsymbol{\epsilon}$, show that $E[\hat{\mathbf{e}}'\hat{\mathbf{e}} \mid \mathbf{X}] = \sigma^2(n - k)$ under homoskedasticity.
4. **Empirically.** Take a random subsample of 200 observations from the GSS extract (`set.seed(1)`) and a design matrix with an intercept, `educ`, and `exper`. Form $\mathbf{P}$ and $\mathbf{M}$ explicitly and verify all the properties above numerically. Report the ten largest values of h_{ii} and describe what those respondents have in common.
5. ★ Show that $\text{Var}(\hat{e}_i) = \sigma^2(1 - h_{ii})$, and explain why this motivates the HC2 and HC3 corrections.

Robust and Clustered Inference (15 points)

1. Derive, for the bivariate model and without assuming homoskedasticity,

$$\text{Var}(\hat{\beta}_1 | X) = \frac{\sum_i d_i^2 \sigma_i^2}{(\sum_i d_i^2)^2}, \quad d_i = X_i - \bar{X}, \quad \sigma_i^2 = \text{Var}(\epsilon_i | X).$$

State the assumption you *do* use about the errors.

2. Explain, in a short paragraph a first-year graduate student could follow, why it is legitimate to substitute $\hat{\epsilon}_i^2$ for σ_i^2 even though $\hat{\epsilon}_i^2$ is a terrible estimate of any individual σ_i^2 .
3. A study has 50 states, 2,000 respondents per state, a treatment that varies only at the state level, and an intra-class correlation of $\rho_e = 0.02$. Using the Moulton factor, by roughly what factor are the naive standard errors too small? A colleague reports $t = 4.1$ from the naive standard error. What is the corrected t ?
4. **Empirically.** For the model `llearn ~ educ + exper + I(exper^2) + female + black`, compute the standard error on `educ` five ways: classical, HC3, clustered on `psu`, clustered on `year`, and clustered on `region`. Present them in a small table. The `region`-clustered standard error is the *smallest* of the five. Explain why that is a red flag rather than a finding, and say which one you would report and why.
5. A referee writes: “The authors should use robust standard errors, since the key regressor is likely endogenous.” Respond in two or three sentences.

Simulation (Optional, ungraded)

Write a simulation in which the true model is $Y_i = 1 + 0.5X_i + \epsilon_i$ with $X_i \sim N(0, 1)$ and $\epsilon_i \sim N(0, (0.5e^{0.8X_i})^2)$, with $n = 200$. Over 5,000 replications, record the coverage of nominal 95% confidence intervals built from (a) classical standard errors, (b) HC0, (c) HC3, and (d) a pairs bootstrap percentile interval. Present the four coverage rates in a table and comment.

A note on effort. Parts 1–3 of each question are algebra and can be done on paper. The empirical parts should take under an hour once you have Lab 1 and Lab 2 open beside you. If you find yourself spending far longer than that, come to office hours rather than grinding.